Sentiment Classification on GoEmotions: A Comparative Study of Text Classification Methods

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***Abstract*—In this work, we study sentiment classification on the GoEmotions dataset by mapping fine-grained emotions to four coarse sentiment categories: positive, negative, neutral and ambiguous. To simplify the task in a single-label setting, only samples with exactly one emotion label are retained. We train and compare two classification models: a Term Frequency-Inverse Document Frequency (TF-IDF) feature pipeline with a Logistic Regression classifier, and a fine-tuned DistilBERT transformer. Our experimental results show that the TF-IDF + Logistic Regression baseline achieves a classification accuracy of 0.636 (95% Confidence Interval (CI) [0.622, 0.650]) and macro-F1 of 0.601 (95% CI [0.585, 0.616]). Fine-tuning DistilBERT on the full training dataset yields a test accuracy of 0.715 (95% CI [0.703, 0.728]) and macro-F1 of 0.684 (95% CI [0.669, 0.699]). Since the 95% confidence intervals for accuracy do not overlap, we conclude that the transformer-based model provides a statistically significant improvement over the linear baseline.**

Introduction

Emotion and sentiment analysis are fundamental tasks in Natural Language Processing (NLP). These tasks enable the conversion of raw, human language into structured data denoting emotional intent, making them useful techniques for a wide range of applications, including customer service, social media analysis, and public opinion monitoring. The widespread adoption of social media has especially led to a massive influx of informal text, which is often characterized by noisy writing, slang, and complex emotional expressions. Standard sentiment analysis (classifying text into positive, negative, or neutral) is often insufficient for capturing the nuances of these diverse expressions.

The GoEmotions dataset (Demszky et al. 2020) is a large, human-annotated corpus with approximately 58,000 Reddit comments labeled with 28 fine-grained emotion categories. In this project, we address a simplified formulation of the task: 4-way single-label sentiment classification. We filter the comments to keep only those with exactly one emotion label, and map that label to one of four sentiment classes: *positive*, *negative*, *neutral*, or *ambiguous.*

To address this task, we compare two modeling approaches. The first is a baseline model that combines TF-IDF features derived from word- and character-level n-grams with a Logistic Regression classifier. The second is a transformer-based approach that fine-tunes a pre-trained DistilBERT model on the training data.

We evaluate both models on the test split, report their classification accuracy and macro-F1 scores along with 95% bootstrap confidence intervals. We also analyze their per-class performance to determine the best-performing model for the task.

Problem Description

The sentiment classification task is framed as a supervised single-label classification problem. Given an input text string $x \in \mathcal{X}$ representing a single Reddit comment, the objective is to predict a label $y \in \mathcal{Y}$. Prior to feature extraction, the text is preprocessed by collapsing consecutive whitespace, removing newline characters, and stripping leading and trailing whitespace. The label space is defined as

$$\mathcal{Y} = \{\text{negative}, \text{neutral}, \text{positive}, \text{ambiguous}\}$$

Sentiment Mapping

We map the original 28 GoEmotions categories to our target sentiment classes as follows:

- **Positive:** admiration, amusement, approval, caring, desire, excitement, gratitude, joy, love, optimism, pride, relief.
- **Negative:** anger, annoyance, disappointment, disapproval, disgust, embarrassment, fear, grief, nervousness, remorse, sadness.
- **Ambiguous:** confusion, curiosity, realization, surprise.
- **Neutral:** neutral.

To formulate the problem as a single-label classification task, we retain only comments with exactly one emotion label.

Methodology

We compare a linear baseline with a transformer-based model for sentiment classification.

TF-IDF + Logistic Regression (Baseline)

The baseline model maps raw comments into a sparse vector space using term frequency-inverse document frequency (TF-IDF) feature weighting. Character-level n-grams are extracted to capture subword details (such as punctuation patterns and word endings), while word-level n-grams capture explicit emotional descriptors. By considering both unigrams and bigrams, the vectorizer captures local lexical patterns, including frequently occurring phrases such as short negation expressions. The resulting vectors are passed to a multi-class Logistic Regression classifier, which models the probability of each sentiment class as a linear combination of the lexical features. To prevent overfitting to the large feature space, the model is trained with L2 regularization, and class weights are adjusted to compensate for training set imbalance.

Fine-Tuned DistilBERT (Transformer)

DistilBERT (Sanh et al. 2019) is a compact transformer model derived from BERT via knowledge distillation to learn deep contextualized text representations. Unlike TF-IDF, which treats words independently of their position, DistilBERT utilizes self-attention mechanisms to dynamically compute representations where a word's vector changes depending on its surrounding tokens. Input text is segmented using WordPiece subword tokenization to mitigate out-of-vocabulary terms. The token sequence is then processed through transformer layers to capture long-range semantic and syntactic dependencies. For classification, the final hidden state corresponding to the special `[CLS]` token is used as a sequence-level representation for classification. A linear classification layer maps this contextual vector to logits corresponding to the target sentiment classes. During training, the pre-trained encoder weights and the classification head are jointly fine-tuned on our target data.

Experimental Setup

This section details the dataset partitions, model hyperparameters, and metrics used in our evaluation.

Dataset Splits

After filtering multi-label comments and mapping the original 28 emotions into four sentiment categories, the resulting dataset is split into training, validation, and test sets. The class counts for each split are:

- **Train Set:** 36,308 samples (12,920 positive, 12,823 neutral, 7,012 negative, 3,553 ambiguous).
- **Validation Set:** 4,548 samples (1,668 positive, 1,592 neutral, 853 negative, 435 ambiguous).
- **Test Set:** 4,590 samples (1,606 neutral, 1,603 positive, 932 negative, 449 ambiguous).

Hyperparameters & Training Settings

For the **TF-IDF + Logistic Regression** baseline, features are computed using word and character n-grams with $n \in \{1, 2\}$. We restrict the vocabulary to the top 50,000 features, filter out features appearing in only one document ($min\_df = 2$), and apply sublinear term-frequency scaling ($\text{tf} \leftarrow 1 + \log(\text{tf})$). The Logistic Regression classifier is regularized with an inverse strength parameter of $C = 1.0$ and optimized using the L-BFGS solver with a maximum of 1,000 iterations.

For **DistilBERT**, input texts are padded or truncated to a maximum sequence length of 64 tokens. The model is fine-tuned for two epochs using the AdamW optimizer with a learning rate of 2×10^{-5} and a batch size of 16. Training was conducted using GPU acceleration.

Evaluation Metrics

We evaluate both models on the held-out test split using:

- **Classification Accuracy:** The percentage of comments predicted correctly.
- **Macro-F1 Score:** The unweighted average of F1-scores across all sentiment classes, ensuring that minority classes (such as "ambiguous") are weighted equally.
- **Per-Class F1-Score:** To evaluate performance variations across specific sentiments.

To quantify uncertainty, we estimate 95% confidence intervals (CI) for both accuracy and macro-F1 using bootstrapping with $N = 1,000$ resamples.

Results and Discussion

Overall Performance

Table 1 summarizes the overall performance of the two models on the test set.

Table I: Overall test performance of the classification models.

Model	Accuracy (95% CI)	Macro-F1 (95% CI)
TF-IDF + LogReg	0.636 [0.622, 0.650]	0.601 [0.585, 0.616]
DistilBERT	**0.715** [0.703, 0.728]	**0.684** [0.669, 0.699]

DistilBERT substantially outperformed the TF-IDF baseline, improving accuracy from 0.636 to 0.715 (+7.9 percentage points) and macro-F1 from 0.601 to 0.684 (+8.3 percentage points). Furthermore, the bootstrap confidence intervals for accuracy show a clear separation between the two models, providing strong evidence that the observed improvement is unlikely to be due to sampling variability.

Per-Class F1 Analysis

Table 2 reports the F1-score for each sentiment class.

Table II: Per-class F1-scores on the test set.

Class	TF-IDF + LogReg	DistilBERT	Δ
negative	0.576	**0.674**	+0.098
neutral	0.629	**0.679**	+0.050
positive	0.747	**0.818**	+0.071
ambiguous	0.452	**0.566**	+0.114

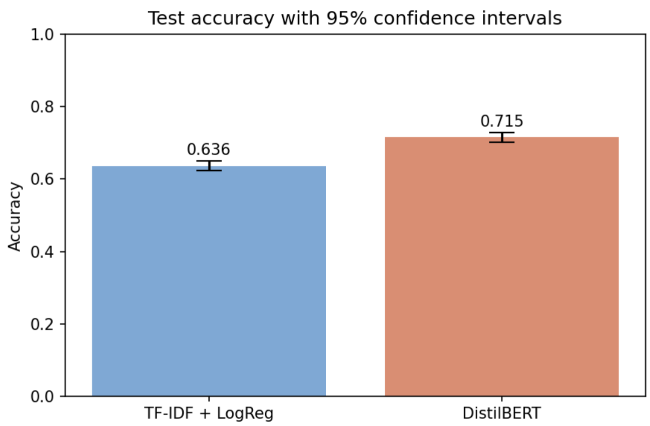

Figure 1: Classification accuracy with 95% bootstrap confidence intervals.

DistilBERT achieved higher F1-scores for every sentiment class. The largest improvements occurred for the **ambiguous** (+0.114) and **negative** (+0.098) classes, suggesting that contextual transformer representations are particularly effective for recognizing indirect, nuanced, and context-dependent emotional expressions. More modest gains were observed for the **positive** (+0.071) and **neutral** (+0.050) classes, indicating that these categories are comparatively easier for both models to identify.

Visualizations

The performance comparisons are illustrated in the figures below. Fig.1 displays the overall test accuracy with 95% bootstrap confidence intervals, Fig.2 compares the per-class F1-scores, and Figs.3–4 present the normalized confusion matrices for both models.

Error Analysis

To understand the limitations of both modeling approaches, we conduct an error analysis on their classification failures on the test split.

TF-IDF + Logistic Regression Errors:

- **Blindness to Valence Shifts and Negation:** Because the model relies on sparse n-gram occurrences, it is blind to contextual shifts in sentiment. For example, it misclassified the negative comment *"I'm sorry to hear that friend :(. It's for the best..."* as *positive* due to the high coefficients of words like `"best"`, `"friend"`, and `"accept"`. Similarly, it misclassified *"It's wonderful because it's awful."* (which is *positive*) as *negative* due to the presence of the word `"awful"`.
- **Over-reliance on Superficial Indicators:** The baseline is highly sensitive to explicit lexical cues. For example, features with large positive-class coefficients, such as `"lol"` (10.28) and `"thanks"` (10.18), strongly influence *positive* predictions, even in sarcastic context. Conversely, since the *neutral* class lacks distinct emotional words, the model is forced to rely on noisy structural features like `"they"` (2.31) or bot commands like `"remindme"` (1.55), which generalize poorly.
- **Out-of-Vocabulary (OOV) and Spelling Variations:** The word-level representation cannot generalize to spelling variations, typos, or slang unseen during training.

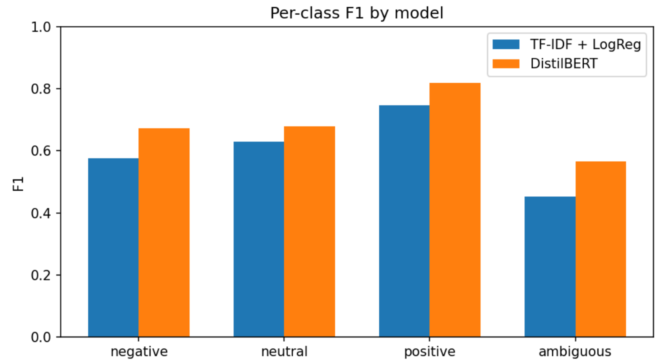

Figure 2: Per-class F1-scores on the test set.

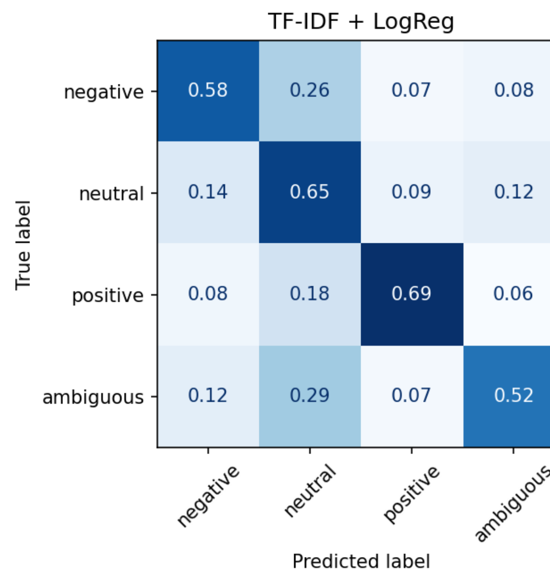

Figure 3: Normalized confusion matrix for TF-IDF + Logistic Regression.

DistilBERT Errors:

- **Sensitivity to Aggressive Phrasing in Neutral**

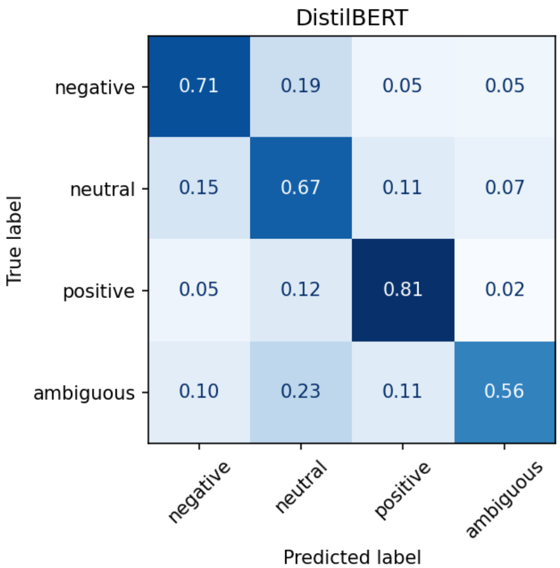

Figure 4: Normalized confusion matrix for fine-tuned DistilBERT.

Comments: DistilBERT tends to over-predict the *negative* class when comments contain strong punctuation or demanding phrasing, even if the content itself is neutral. For example, it misclassified *"You take that back right now! [NAME] does not deserve this."* (labeled *neutral* in GoEmotions) as *negative*.

- **Failure on Short, Elliptical Queries:** For short, elliptical queries, DistilBERT may struggle when structural cues dominate the meaning of the sentence. For instance, for the ambiguous question *"out of everyone why AC?"*, DistilBERT predicted *neutral*, whereas the baseline correctly identified it as *ambiguous* by matching the key feature `"why"`.
- **Interpretability Deficit:** Unlike the logistic regression baseline where the feature coefficients directly explain why a classification was made, DistilBERT's deep representations act as a black box, making individual failures difficult to debug or audit.

Conclusion

These findings demonstrate that contextual transformer models substantially improve sentiment classification of noisy social media text. While TF-IDF remains valuable when efficiency and interpretability are priorities, DistilBERT provides a clear advantage for modeling context-dependent emotional expressions.

Demszky, Dorottya, Dana Movshovitz-Attias, Jeongwoo Ko, Alan Cowen, Gaurav Nemade, and Sujith Ravi. 2020. "GoEmotions: A Dataset of Fine-Grained Emotions." In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 4040–54. Online: Association for Computational Linguistics. https://doi.org/10.18653/v1/2020.acl-main.372.

Sanh, Victor, Lysandre Debut, Julien Chaumond, and Thomas Wolf. 2019. "DistilBERT, a Distilled Version of BERT: Smaller, Faster, Cheaper and Lighter." *arXiv Preprint arXiv:1910.01108*.